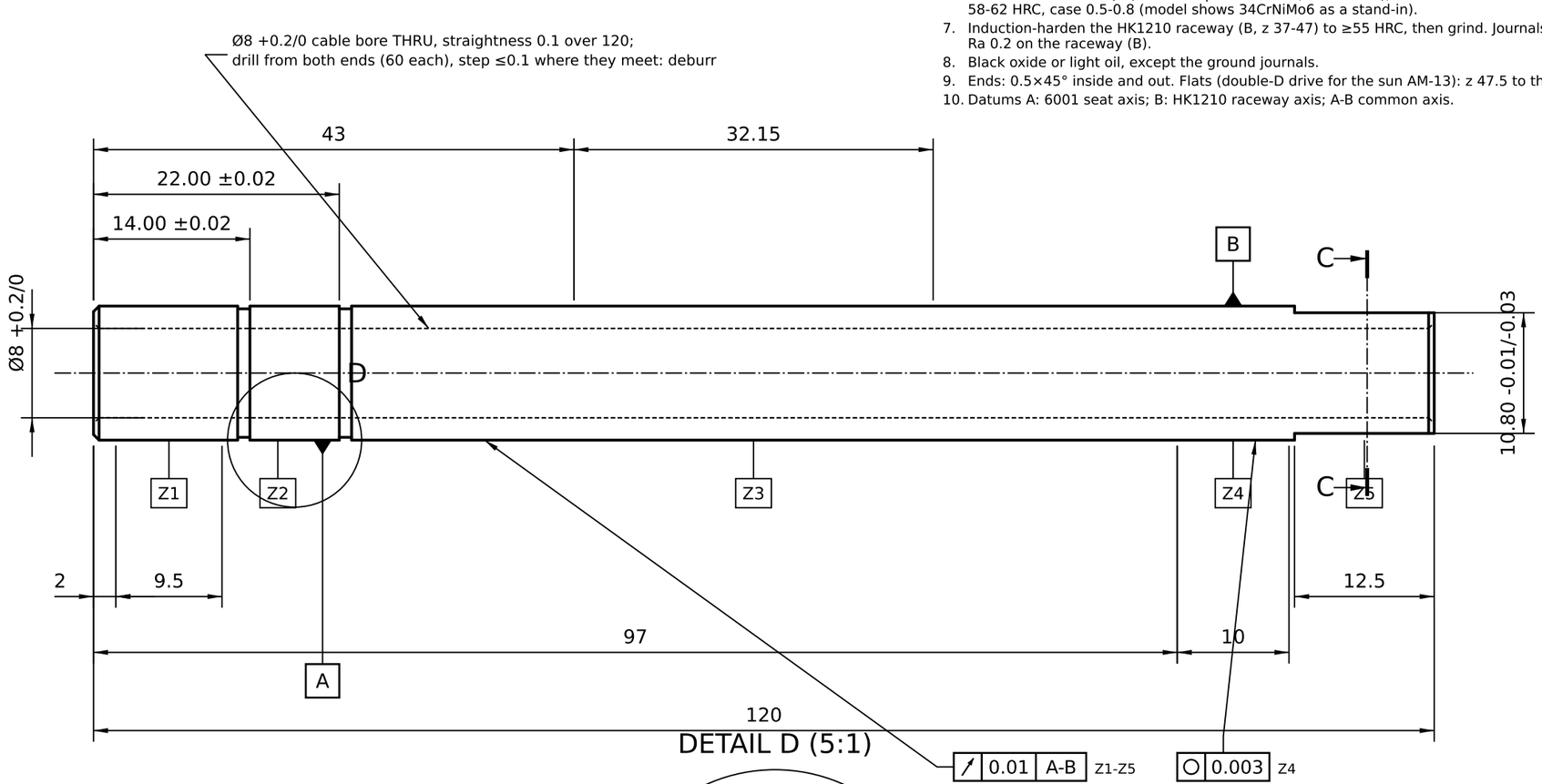
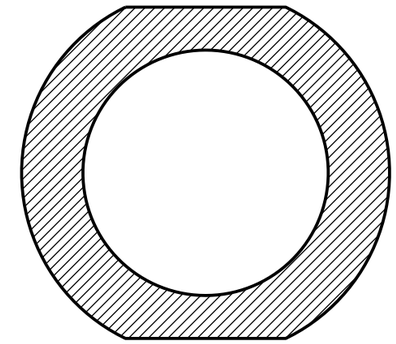
MOTOR SHAFT (2:1)

NOTES

1. Dimensions in mm. General tolerances ISO 2768-mK. Fits ISO 286-1/-2 (deviations shown in brackets).
2. STEP file governs undimensioned geometry: cnc/step/rotor.step (STEVE 02e31a10 rev 48).
3. Edges ISO 13715 -0.3 (break 0.2-0.4 x 45°) unless a chamfer is shown. Remove all burrs, especially in bores and holes.
4. Surface roughness Ra 3.2 µm unless noted (ISO 21920-1).
5. Threads ISO metric coarse, 6H internal / 6g external; thread depths are full-thread depths.
6. Material 42CrMo4 +QT (AISI 4140 pre-hardened, 28-32 HRC), with certificate. Alternative 16MnCr5 / 8620 carburised 58-62 HRC, case 0.5-0.8 (model shows 34CrNiMo6 as a stand-in).
7. Induction-harden the HK1210 raceway (B, z 37-47) to ≥55 HRC, then grind. Journals ground. Ra 0.4 on the 6001 seat (A), Ra 0.2 on the raceway (B).
8. Black oxide or light oil, except the ground journals.
9. Ends: 0.5×45° inside and out. Flats (double-D drive for the sun AM-13): z 47.5 to the front end.
10. Datums A: 6001 seat axis; B: HK1210 raceway axis; A-B common axis.



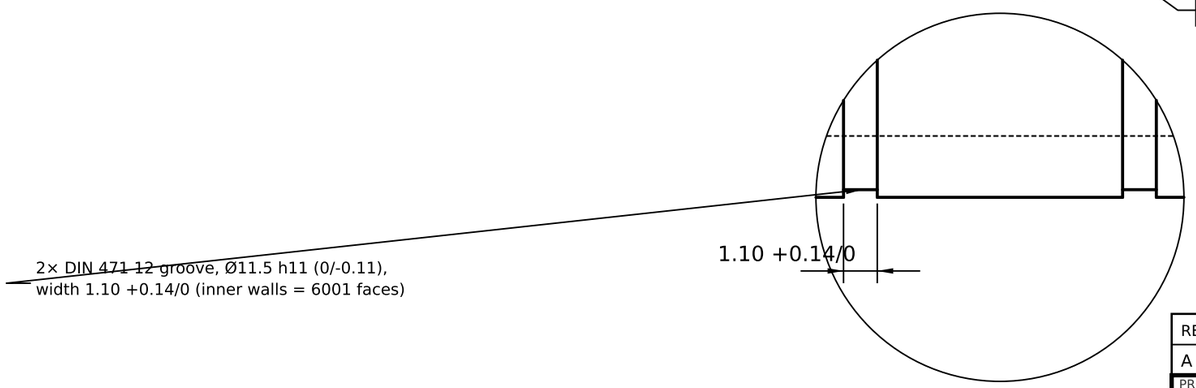
SECTION C-C (5:1)



0.02 A-B

flats 10.80 -0.01/-0.03 across, symmetric to A-B within 0.02, parallel to each other 0.01

DETAIL D (5:1)



ZONE	FROM REAR END	Ø12	Ra	FUNCTION
Z1	2.00 - 11.50	g6 (-0.006/-0.017)	3.2	encoder hub AM-11 (Loctite 638)
Z2	14.00 - 22.00	k5 (+0.009/+0.001)	0.4	6001 inner ring; datum A
Z3	43.00 - 75.15	p6 (+0.029/+0.018)	3.2	hub sleeve AM-10 press fit
Z4	97.00 - 107.00	h6 (0/-0.011)	0.2	HK1210 raceway ≥55 HRC, circularity 0.003; datum B
Z5	107.50 - 120.00	g6 (-0.006/-0.017)	3.2	sun AM-13: round portions of the double-D

Ø12 outside the zones: ISO 2768-m. Zones Z2 and Z4 ground; runout of Z1-Z5 to A-B 0.01.

REV	DESCRIPTION	DATE	BY
A	First release for quotation and manufacture	2026-10-14	Claude

PROJECT: **Unknown machine actuator**	DRAWN BY: Claude for macncrash	SOURCE (MODEL GOVERNS): STEVE 02e31a10 rev 48; STEP: cnc/step/rotor.step
GENERAL TOLERANCES: ISO 2768-mK (ISO 22081); Edges ISO 13715 -0.3 (0.2-0.4)	PROJECTION: FIRST ANGLE; ISO 128 / ISO 5456-2	
FINISH / HEAT TREATMENT: Induction-harden raceway ≥55 HRC, ground; black oxide		UNITS: mm
MATERIAL: 42CrMo4 +QT (4140 HT)		SCALE: 2:1 (views as noted)
TITLE: **Motor shaft (hollow)**		QTY PER ACTUATOR: **1**
DRAWING / PART NUMBER: **AM-09-motor_shaft**	REV: **A**	SHEET: 1/1 · SIZE: A3 · DATE OF ISSUE: 2026-10-14